

Timestamp & Validation-Based Protocols DBMS

> **Definition:** Timestamp Ordering Protocols ensure conflict serializability without using locks by assigning a unique monotonic timestamp **TS(T_i)** to each transaction upon start and enforcing operation ordering according to timestamp sequence.

1. Detailed Technical Explanation

1. System Timestamps

Each data item **X** maintains two timestamps:

2. Basic Timestamp Ordering Rules

Read Operation `Read(X)` by Transaction **T_i**:

Write Operation `Write(X)` by Transaction **T_i**:

2. Validation-Based (Optimistic) Protocol

Used when read operations dominate write operations (low conflict rate). Execution is split into 3 phases:

Validation Test Condition:

For all **T_k** committed before **T_i**:

3. Core Concepts & Memory Keywords

- **Monotonic Timestamps:** Unique growing values (System clock or global counter).

4. Must-Write Points for Exams

- Timestamp ordering protocols guarantee **freedom from deadlocks** because transactions are never made to wait indefinitely.

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5. Quick Recall Flow

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